Utilizing Modern Deep-Learning Techniques in Mass Spectrometry Bacterial Classifiers

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***Abstract*—This work focuses on applying modern Machine Learning and Deep Learning technologies to the existing domains of mass spectrometry and bacterial identification. Our work performs a comparative analysis of an attention-based model, a convolutional neural network, and a basic feed forward network. We compare a number of architectures for each model and identify a basic feed forward network as performing the best. We then provide analysis and investigation on the struggles faced by the more complicated deep learning technologies and suggest opportunities to improve on this base level performance.**

***Index Terms*—Deep learning, Convolutional neural networks, Neural Network**

I. Introduction and Motivation

Clinical microbiology laboratories and hospitals are continuously seeking to reduce the time and resources required for diagnosis. Matrix-assisted laser desorption/ionization time-of-flight mass spectrometry (MALDI-TOF MS) has been widely incorporated in clinical laboratories due to its high accuracy and ability to identify microbial isolates within minutes, compared to the 24 to 48 hours required by other conventional methods [1]. MALDI-TOF MS technology identifies microbes by producing mass-to-charge (m/z) peaks from samples and comparing them with reference spectra stored in databases [1, 2]. Although this reference-matching method is rapid, identification reliability may be compromised by variations in sample preparation, bacterial growth conditions, and other non-standardized laboratory factors [1].

To address these limitations, traditional machine learning classification methods such as k-Nearest Neighbors (KNN), Support Vector Machines (SVM), or Random Forests (RF) have been applied to MALDI-TOF data as demonstrated by Mahé et al. and Mortier T et al., to enhance discrimination between species [2, 3]. More recently, deep learning algorithms such as convolutional neural networks (CNNs) have been studied for similar purposes [3, 4]. With this project, we propose to address microbial identification as a supervised classification problem, using MALDI-TOF mass spectra to classify samples into one of 20 bacterial species by applying modern deep-learning models to improve robustness in microbial identification. To test the hypothesis that modern models could improve microbial identification, we develop several models and test their performance against previous results guided by the research question: can modern deep-learning models as CNNs and Attention-based models improve the robustness of microbial identification using mass spectrometry (MALDI-TOF) data?

II. Related Work

Mahé et al introduces microbial classification from MALDI-TOF MS as a long-standing classification task with significant implications for the clinical identification of microbes in a fast and affordable manner [2]. They go on to present a method for classifying MALDI-TOF MS data using a penalized non-negative linear regression framework, with which they achieved 97.1% correct identification for pure spectra (spectra generated with only one genera of microbe in the sample). Notably, one purpose of this study was to also classify spectra from mixed samples containing multiple genera of microbes. For this reason, their model was constructed to predict the presence of multiple genera based on the observed spectra. While this generalizability of the model is a strength, the model's performance on pure spectra could likely be improved by using deep learning techniques, which are absent from their approach.

More recent studies have addressed the lack of deep learning techniques in MALDI-TOF analysis. For example, Mortier et al. tested a variety of machine learning techniques on a large

data set of almost 100,000 spectra and over 1,000 species [3]. They found that a 1-dimensional convolutional neural network with ReLU activation and Monte Carlo dropout was unable to outperform classical learning models, such as KNN, in detecting novel strains of bacteria represented in the training data, but was much more effective than classical learning models at identifying novel species. These findings demonstrate the efficacy of CNN models in learning to differentiate bacterial samples at the species level, but further architectural tuning is necessary for creating an effective predictor for smaller datasets such as the one utilized by Mahé et al [2]. Towards this end, Seddiki et al. have explored the efficacy of transfer learning when applied to MALDI MS data [4]. By fine-tuning a 1-dimensional convolutional neural network trained on rat brain tissue samples, they achieved enhanced performance on a variety of classification tasks based on small datasets. This is a promising result, but the models began with raw spectra, and the benefits observed in this paper may not apply to peak-list data, which is the data type on which this current study is focused.

III. Data & Analysis

Our data set consists of MALDI-TOF MS data collected from 20 Gram positive and negative bacterial species. There are 651 instances of pure spectra data that will be the focus of this study. The MS spectra were obtained via a standard culture-based workflow, in which the bacteria was first grown on an agar plate for 24 to 48 hours, before a portion of colony was picked, spotted on a MALDI slide, and a mass spectrum was acquired. The MS spectra were represented in the peak-list format used by the commercial VITEK-MS system. Therefore, each data sample consists of a list of signal intensity values at peaks for mass-to-charge ratios ranging from 0 to 1300 daltons/charge. The dataset is available in the UCI Machine Learning Repository under the name "MicroMass" [5]. The baseline classical technique established in Mahé et al, supported an accuracy of 97.1% can be used as a benchmark for the models developed in this project.

Figure 1 diagrams the relationship between bacterial species as identified in Mahé et al, with more closely related species having a closer relationship.

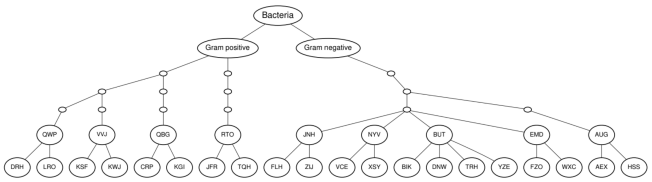

Fig. 1. Relation of bacterial species from Mahé et al [2].

Figure 2 outlines the class distribution of the bacteria species. The class distribution suggests a unbalanced dataset. This is also true within near species relationships where some closely related species have well over double the number of samples of similar related classes, e.g. there are 67 samples of "BUT.YZE" but only 18 of "BUT.BIK". To help address this issue stratified sampling was used to ensure that all classes were represented amongst the train, validation, and testing sets.

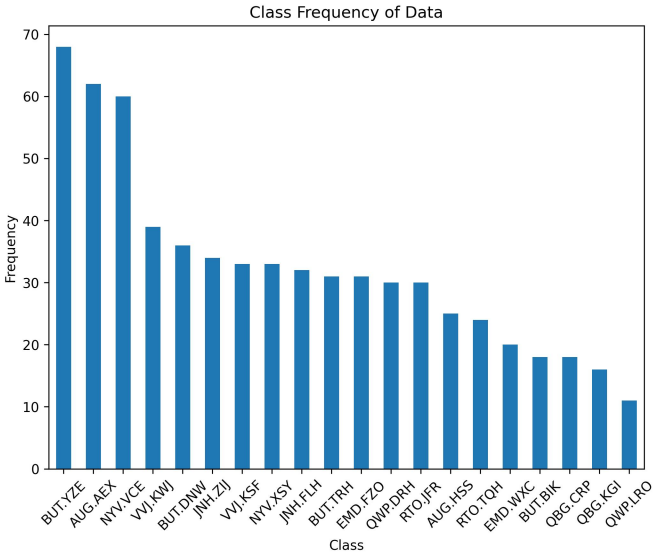

Fig. 2. Class Distribution of Data Set

An example sample of the bacteria Haemophilus Influenzae can be seen in Figure 3. The scale is on the scale of 10^6 arbitrary units with the maximum value of approximately $4.8 * 10^6$. Data of this scale presents a challenge to deep learning technologies where gradients are calculated as a multiplication of the input data. The exponential nature of the data also causes loss of fidelity between mass charge bins, where information is lost in noise to exponentially larger values.

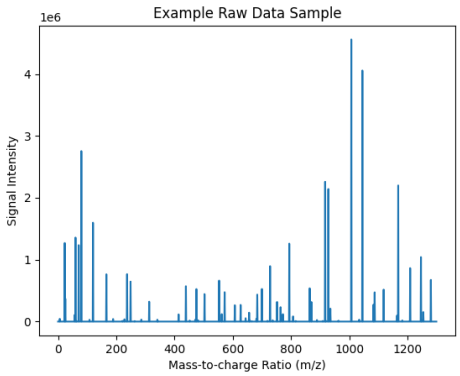

Fig. 3. Mass spectrometry profile of sample of Haemophilus Influenzae

To address these issues log1p scaling was used as a pre-processing step which significantly reduced the scale, as well as balanced the size of smaller valued features. log1p was selected as all of the values are positive greater than or equal to 0 and log1p ensures that all of the values in this domain exist by adding 1 the value before taking the natural logarithm.

Logarithm was selected to improve fidelity between mass-charge ratio bins of exponentially different sizes. The sample from Figure 3 after preprocessing can be seen in Figure 4.

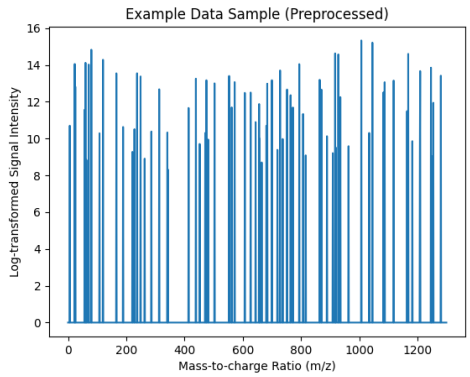

Fig. 4. Mass spectrometry profile of Haemophilus Influenzae after preprocessing is applied.

IV. Methodology

To capitalize on the improvements of modern machine learning technologies over the classical approach taken by Mahé et al, we utilized three modern deep-learning technologies: feedforward networks, convolutional networks, and attention mechanisms. Convolutional neural networks and attention model architectures were selected for their relevance as high-performing architectures capable of modeling relationships between the spectra peaks with spatial context. For each of these, we constructed and validated a number of different network architectures, both with Leaky ReLU and standard ReLU activation functions.

Stratified random sampling on the species class was used to split mass spectrometry data into a set for training and validation, consisting of 90% of the data, and the remaining 10% was reserved as a test set. All of the models were then evaluated using a 5-fold cross-validation for each variation of architecture, and final evaluation was performed on the model with the highest mean cross-validation accuracy.

Once all model architectures were trained and evaluated with 5-fold cross-validation, the model with the highest mean cross-validation accuracy was identified, retrained on the total training and validation set, and evaluated on the held-out test data.

V. Results

A. Feedforward Neural Network Baseline

A number of feed-forward networks were trained. They consisted of a series of layers each followed by a dropout layer of 20%. Each architecture was also tested using both normal ReLU and Leaky ReLU activation functions. The following architectures were tested to tune hyper parameters for the basic model and create a well performing baseline.

Layer 1 Width	Layer 2 Width	Layer 3 Width	Parameters
256	256	None	407612
1024	128	None	1470908
1024	256	None	1604924
1024	256	128	1635516

All of the the above architectures were trained using both traditional ReLU and leaky ReLU. The cross-validation results of each variation of the model can be seen in Figure 5. The best overall performance was with traditional ReLU and two layers of size 1024 and 256 at an average cross-validation accuracy of 0.950. This provides a consistent baseline for the more complicated network architectures.

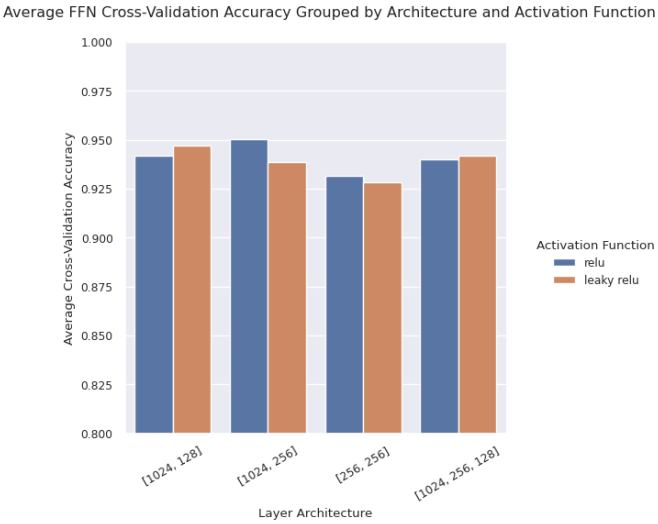

Fig. 5. Performance of different feed forward networks with hidden layer sizes. Numbers in brackets correspond to width of each feed-forward layer.

B. Convolutional Neural Network

Convolutional neural networks provide a good theoretical model for mass spectrometry data in 1 dimension. Convolutional neural networks learn patterns across the spatial dimension and embed it in feature dimensions. Mass spectrometry data is spatially organized with feature patterns locally and on the scale of the entire data set contributing to the full classification. Specifically, each peak in the spectra corresponds to a charged portion of a molecule. The peaks are then reassembled into larger biomolecular features that are indicative of the bacterial species in the sample.

For the convolutional neural network a number of architectures and hyperparameters were explored. Each one consisting of interleaved convolutions with max pooling layers before being flattened and connected through a 128 with ReLU activation. These were each compared with both ReLU and Leaky ReLU activations for the convolutional layers. Architectures consisting of the following hyperparameters were explored.

Layer 1	Layer 2	Parameters
32 filters, 32 size kernel	32 filters, 64 size kernel	2733942
32 filters, 16 size kernel	32 filters, 64 size kernel	2733430
32 filters, 32 size kernel	64 filters, 32 size kernel	1400598
32 filters, 16 size kernel	64 filters, 32 size kernel	1400086

We achieved the best performance with a CNN (convolutional neural network) with Leaky ReLU activation, an initial convolutional layer with kernel size of 32 and 32 filters, and a second convolutional layer with a kernel size of 32 and 64 filters. Our optimized CNN achieved an average cross-validation accuracy of 0.942.

The comparison between the trained CNN architectures can be seen in Figure 6. All of the evaluated convolutional neural network approaches performance between 0.92 and 0.942, with the best being at 0.942.

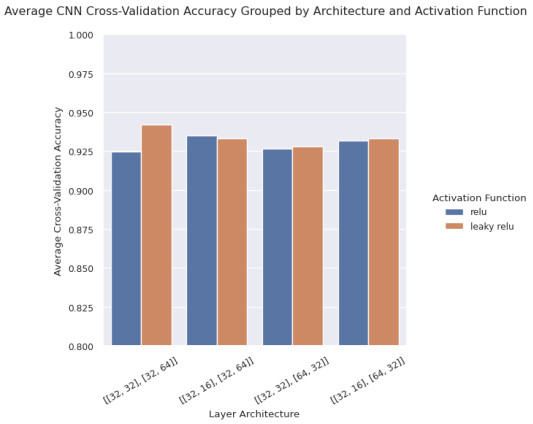

Fig. 6. Performance of different convolutional neural networks. Each pair of numbers in nested brackets corresponds to the number of filters and kernel size for each layer.

C. Attention Based Neural Network

Attention was also investigated for its ability for specific spectra peaks to attend to other peak data allowing for relationships across the spectral range. These relationships between multiple peaks are informative of the biomolecular features that help distinguish one bacterial species from another. The theoretical benefit would be that the model is capable of learning which peaks have information that combine to provide informative information regarding the final classification of the spectra.

Attention also has a benefit over the more traditional convolutional neural network in that the traditional attention mechanism is capable of seeing the entire input and attending all spectra peaks to all other spectra peaks. Unlike the convolutional neural network, which has a more limited receptive field, the attention architecture sees the entire input. Our attention mechanism also includes a basic skip connection, ensuring that the input information isn't lost through the attention layer. For positional encoding in our network we used a sinusoidal position layer that appended position vectors.

Architectures with 8 and 16 heads and 8 and 16 for the key dimension were tested. Both of these were run together to test hyperparameters of the following.

Number of Heads	Key Dimension	Parameters
8	8	6690198
8	16	6756374
16	8	6756374
16	16	6987030

The performance of the many different attention-based architectures can be seen in Figure 4. Nearly all of the architectures performed in the range of 0.9 to 0.925 average cross-validation accuracy, peaking at 0.925, for the model with a leaky ReLU activation function, 8 heads, and a key dimension of 16.

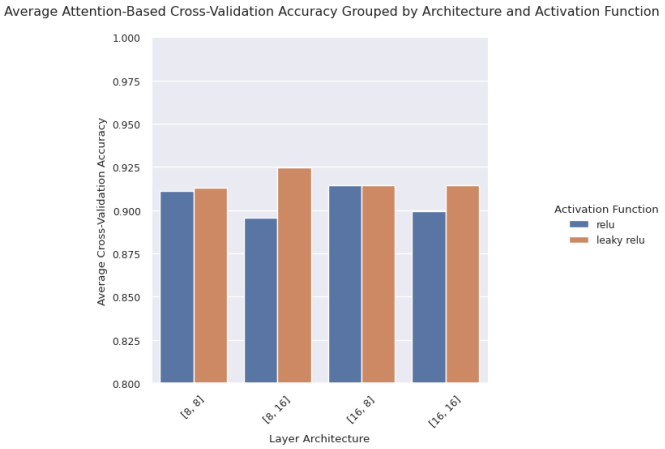

Fig. 7. Performance of different attention networks. The numbers in brackets correspond to the number of heads and key dimension.

D. Inter-model analysis

All of the models achieved fairly strong cross-validation performance, but the feed-forward network performed slightly better than any of the convolutional neural networks, and attention networks performed the worst. The overall best model was identified as the feed-forward architecture with a ReLU activation function, and two layers of size 1024 and 256 at an average cross-validation accuracy of 0.950. The best performance of the optimal architecture for each model type can be seen in Table 1.

Network Type	Validation Accuracy
Feed Forward Neural Network	0.950427
Convolutional Neural Network	0.941880
Attention Based Neural Network	0.924786

TABLE I
INTER-MODEL ANALYSIS BEST RESULTS

The best performing model of the feed forward network with two layers, one of 1024 nodes then one of 256 nodes was then trained on the entirety of the test-train dataset. This was then evaluated on the test set with an accuracy of 0.924.

VI. Discussion

A. Interpretation of Results

The simple feed-forward network achieved the best results. This is likely due to some of the limitations of the more

complicated architectures investigated. Both the CNN and Transformer based architectures are significantly more difficult to train due to their number of parameters. This makes it more likely that the networks will learn noise or memorize the training data. This issue is expounded by the relatively small dataset that the model is trained on 651 instances. This is a relatively small amount of data to train these complex models on from random initializations.

To be able to resolve this issue a larger amount of data would help support learning in more complicated models. Further greater data augmentation techniques would be able to help improve data diversity supporting generalization of the limited training dataset.

B. Ad Hoc Analysis

Figure 8 shows the per class accuracy of the final model on the test set. It is notable that the worst performing classes were those with high imbalances between similar species this is seen in the complete failure of the final model to accurately identify any of the class ”QWP.LRO” class and the struggling performance of the model on the minority classes for ”BUT.DNW” and ”BUT.TRH” compared to the majority class ”BUT.BIK”, as well as ”NYV.XSY” vs ”NYV.VCE”. This result suggests that there is a strong bias in the model resulting from similar species that have signficantly varying frequencies.

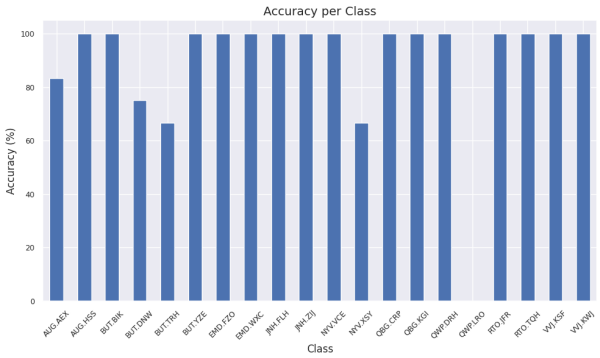

Fig. 8. Test accuracy by class of data

Notably as well the accuracy of the model on the test set regarding smaller classes such as ”QWP.LRO” is subject to the relatively small initial data size the stratified train-test split. Meaning that there are relatively few samples in the smaller classes and the confidence on these smaller frequency test classes is less certain.

Further with classes with only a few samples it becomes progressively more difficult for the model to learn generalizable patterns in the data.

VII. Conclusion

This investigation looks into the power of using modern deep learning in identifying minute differences between similar mass spectrometry data from hard-to-differentiate microorganisms. In this project, we evaluated the efficacy of three different forms of deep learning technologies – a feed-forward neural network, a convolutional neural network, and an attention-based neural network – and selected the optimal architecture for our classification task, based on 5-fold cross-validation results. Thus far, our work expands the results of previous research by Mortier et al. [2], who demonstrated initial success in using CNNs for mass spectrometry data, by investigating how other techniques can be used. However, we ran into issues regarding more complicated machine learning technologies as a result of our data. Our work shows that advanced ML techniques for sparse data, such as leaky ReLU activation, improved model performance for classification tasks in MALDI-TOF MS, as models using Leaky ReLU consistently outperformed standard ReLU across our tested architectures and in our selected optimal model. However using more complicated techniques is heavily dependent on the amount of data that is available.

VIII. Future Work

Future work may look at developing further techniques in data augmentation for MALDI-TOF for using modern deep learning techniques on limited data. As well as methods of better balancing class distributions such as resampling or class weighting.

References

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Self-Declaration

Self Delcaration: We, the undersigned, confirm this proposal is a group project. Each member contributed to the planning and implementation strategies.

Division of tasks:

- **Connor Donhue:** Wrote Introduction and Related Work. Finalized notebook code.
- **Connor Fleser:** Wrote initial versions of the code, directed model selection, wrote Data & Analysis, Results, part of Methodology section, and supported revisions.
- **Jessica Pamela Feliz Garrido:** Wrote Conclusion and Future Work. Supported revisions, managed the references and project organization.
- **Matthew Harrelson:** Wrote parts of most sections in the report, including doing large amounts of revisions and copy-editing. Ensured organized formatting in LaTeX.